Jian Feng

j36feng@uwaterloo.ca | 647-766-0178 | Website | linkedin/in/fenjian | github/faketut

EDUCATION

University of Waterloo, Computing and Financial Management, Honours, Co-op 09/2023 - 04/2028
Majors: Computer Science, Finance, Statistics

EXPERIENCE

Manulife 05/2026 – 08/2026
Co-op
- Triage ~10 weekly cloud and network infrastructure tickets from global ETS teams via ServiceNow, owning resolution and stakeholder follow-through.
- Co-developing `aks-onboarding-automation` to replace manual ServiceNow ticket exchanges, projected to compress onboarding turnaround from days to minutes per request.

Treasury Board of Canada Secretariat 09/2025 – 12/2025
Co-op
- Shipped a .NET MVC module that auto-normalized job titles across 5K public firms using pattern-matching heuristics; reduced manual data cleaning by 80%.
- Migrated 30 test cases across 5 suites from legacy Selenium to Katalon and stood up CI/CD on Azure DevOps, presenting demo to executives.
- Refactored core modules of an enterprise data platform and optimized SQL queries, cutting page load times by 30% while improving long-term maintainability.

Lansi Institute of Artificial Intelligence in Medicine 04/2024 – 12/2024
Co-op (Remote)
- Built genomic ETL pipelines for multi-omics data (~100+ GB, FASTQ/VCF), orchestrating per-sample fan-out across cores with Python `multiprocessing` to compress end-to-end processing 89%.
- Co-authored a published T2D drug-target meta-analysis pooling 68 studies; trained XGBoost, SMR, and MR^2 models to link genetic variants with drug response.

Sanyang Technology Co., Ltd. 07/2024 – 11/2024
Intern
- Built foundational components of a flexible job-shop scheduling (FJSP) system, enabling the algorithm team to run macro production-planning experiments end-to-end.
- Delivered CRUD APIs across multiple cloud-platform product modules, unblocking upstream and downstream system integration on schedule.

PROJECTS

ArmyOfSafeguards
- Ensemble of four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable through env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold/weighted baselines on held-out HF benchmarks.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

GhostPilot
- Invisible Windows overlay streams WASAPI loopback audio through partial/final Azure ASR into a text LLM, with a watchdog auto-restarting the capture pipeline.
- Alt+P vision path: Qt area capture → JPEG → vision LLM, result folded back into the text-LLM prompt.
- Local RAG over `knowledge/` injects user-curated context so answers stay grounded across long sessions.

MiniKV
- Go LSM engine—CRC32C WAL, skip-list MemTable, snappy block SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable optimistic transactions: snapshot reads, buffered writes, commit-time read-set validation against a bounded commit log—atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC streaming; followers resume from local `LastSeq()` on reconnect so applied events aren't replayed.

Certificates & Skills

Coursera: Duke Data Engineering, IBM Vector Database, UCI Database Design & Operational BI, IBM AI for Cybersecurity, Standford Machine Learning, UCI Mathematics for ML, UCSC Bayesian Statistics, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning **Skills**: Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, Spring Boot, React, Node.js, Express.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, Pytorch, TensorFlow, Scikit-learn, OpenCV, GenAI tools